AI Learning Assistant:

Technical Approach plan,

1. Objective

Build a conversational AI chat assistant for students in Kenyan education environment to understand their class and subject material through Natural language interaction.

The system allows student to ask questions through text and where applicable images of textbook problem or handwritten work.

It should not just generate answer but provides a response that is:

1. Correct and Easy to understand
2. Appropriate for the student's level.
3. Consistent across conversations.
4. Honest when system does not have enough info.
5. Available under real school connectivity constraints.

The initial objective is to validate the core experience with a focused curriculum before expanding into personalization, adaptive learning and other advanced capabilities.

2. Kenya Context

The system should be designed around the actual Kenyan curriculum and school environment.

The knowledge base should be structured as:

Grade/Class → Subject → Chapter/Topic → Content

Each curriculum item should carry metadata such as:

- Grade/Class
- Subject
- Topic/Chapter
- Curriculum
- Language
- Difficulty where available

Language

Model selection cannot be based only on general benchmarks.

The evaluation should have separate:

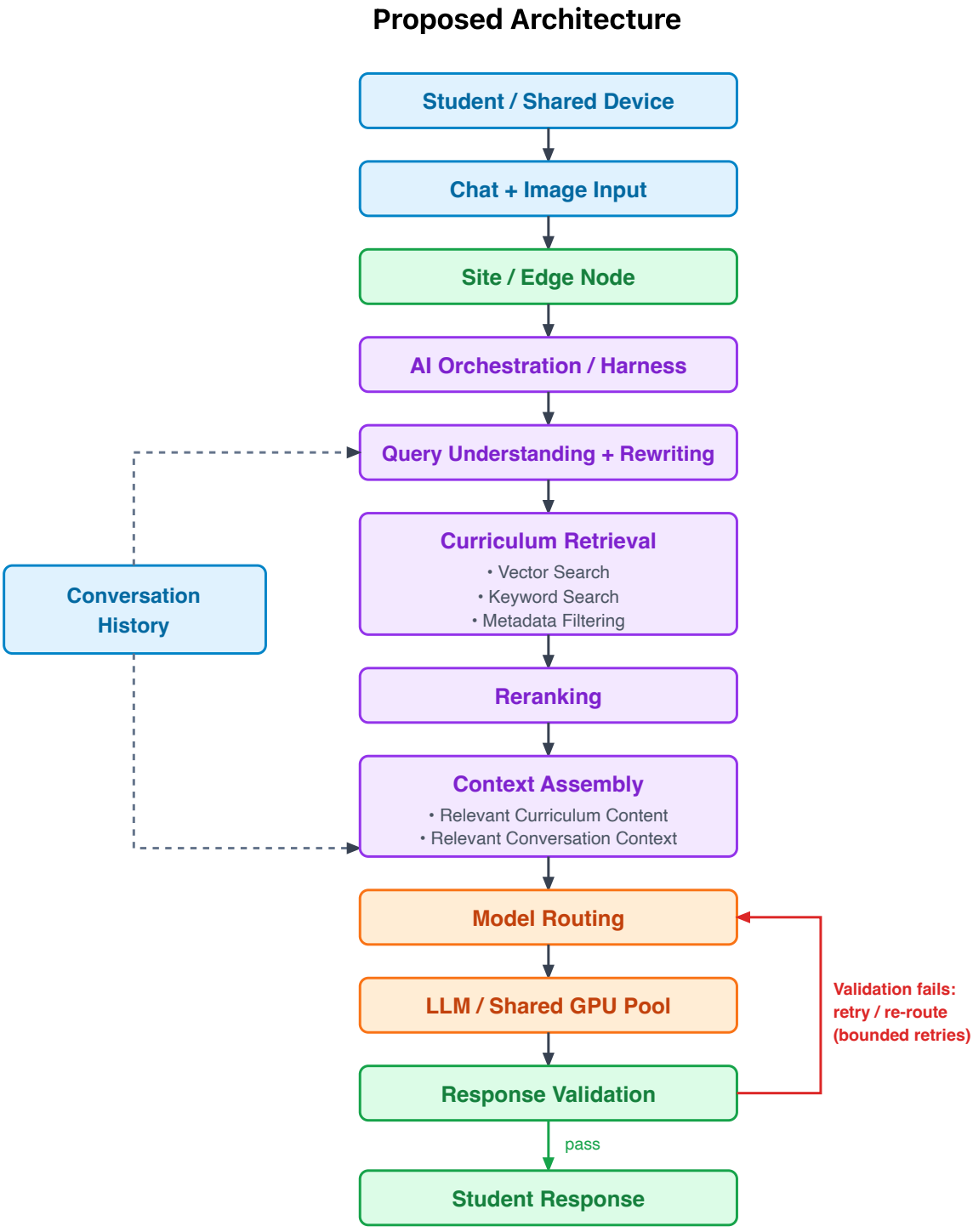
- English track
- Kiswahili track

Model candidates such as Qwen3-8B, Qwen3-14B and Mistral should be evaluated against actual Kenyan educational questions in both languages.

The final model should be selected based on:

Educational Quality × Latency × Inference Cost

3. Architecture



4.Query Understanding and conversation context

Conversation context should not bypass curriculum retrieval.

Example,

Student: What is a fraction?

And follow up with,

Student: Can you give an example?

Students may ask short follow-up questions such as “Why?”, “How does that work?” or “Can you give another example?” These questions may not make sense when treated independently.

The orchestration layer should use the current question and relevant conversation history to rewrite the question into a form that can be searched effectively.

5.Curriculum grounded RAG

The knowledge layer will use hybrid retrieval

- Vector search for semantic relevance
- keyword search
- Metadata filtering for class/subject/topic
- Reranking before sending context to LLM

This keeps the Model grounded and help in reducing hallucinations.

A PostgreSQL-based setup with vector search is one option for the knowledge store. The final retrieval configuration should be selected based on retrieval quality and operational requirements.

6.AI Orchestration / Harness

Rather than treating the LLM as the entire application, I would build a controlled AI orchestration/harness layer around the model.

The LLM will not operate in isolation, A lightweight orchestration layer will control the following things:

- Query Understanding.
- Query rewriting
- Context selection.
- Prompt construction
- Model selection
- Response Validation
- Token limits
- Observability

For selected failures:

Generate → Validate → Correct → Respond

The correction loop will be strictly bounded to prevent unnecessary latency and inference cost

7.Response validation

Validation should balance reliability against latency and inference cost

Rather than using LLM judge for every response, validation can used in multiple layers:

Layer1: Basic Checks,
Fast checks such as required context, response, format and basic safety constraints.

Layer2: Grounding/quality checks,
check whether the response is supported by the retrieved curriculum context and meets expected answer characteristics.

Layer3: Stronger validation when required,
Use a light weight classifier or model-based check for responses that are uncertain, complex or higher risk

The system should measure the latency of validation and avoid blocking streaming unnecessarily

8.Image Input

Image input should be supported as an additional input path.

Examples include:

- Photo of a textbook problem
- Handwritten mathematics
- Diagram based questions

And the flow would be:

Image → OCR / Vision Understanding → Structured Question → Normal Query + RAG Pipeline

This keeps image understanding separate from the core curriculum retrieval architecture while allowing both text and image questions to use the same reasoning pipeline.

9.Deployment and Connectivity

The target deployment should reflect the proposed hybrid model

Site edge nodes + Shared GPU inference Pool

Normal connectivity
Student request → Site/Edge → Shared GPU pool → Response

Low bandwidth,
Reduce unnecessary payload and context transfer and use locally cached curriculum/session information where possible.

Temporary connectivity loss,
The active session should not be lost. Session state and non-critical events can be synchronized when connectivity returns.

Shared devices,
The system should support multiple learners using shared lab devices while maintaining isolated sessions and learner context.

Offline behaviour,
Define offline capabilities based on the inference resources and curriculum content available on each edge node. The system should degrade gracefully rather than imply full AI functionality without connectivity..

10.Modal Strategy

I would avoid committing to a single model before testing the actual educational workload.

Model	Purpose
Qwen3-8B	Initial efficiency baseline
Qwen3-14B	Quality benchmark
Mistral Small 3.1 24B	Alternative quality benchmark

The models should be evaluated on:

- Kenyan curriculum questions
- English
- Kiswahili
- Simple questions
- Multi-step reasoning
- Follow-up questions
- Curriculum-grounded questions
- Out-of-scope questions
- Image-derived questions where applicable

The final model should not be selected upfront. It should be selected from measured educational quality, latency, throughput and inference cost

Also, the system can eventually use model routing like smaller/cheaper models for simple questions and stronger models for complex reasoning.

11.Scale and Capacity Model

The initial deployment context is:

- 30 sites
- 38400 approx. learners

The key workload variables are:

- Learners per site
- Daily active learners
- Questions per learner per day
- Average input tokens
- Average output tokens
- Retrieval context size
- Peak concurrent sessions
- School-hour traffic distribution
- Target response latency
- GPU tokens/second
- Model quantization/configuration

Parameters	Planning assumption
Total learners	38,400
Daily active learners	25%
Questions per active learner per day	3
Estimated questions per day	28,800
Average combined input and output tokens per question	900
Estimated total tokens per day	25.9 million

These are planning assumptions, not production measurements. Actual usage should replace them once pilot data is available.

GPU count should then be derived from measured model throughput and the required peak concurrency.

The resulting model should report:

GPU requirement → monthly inference cost → cost/question → cost/learner/month

with low/base/high usage scenarios.

12.Cost and Scalability principles

Cost should be considered form architecture stage.

The optimization approaches:

- Use the smallest model that meets quality requirements
- Route complex questions to large model only when necessary
- Keep retrieved context focused
- Summarize long conversations
- Cache reusable retrieval or results
- Strictly limit retry loops

The important production metrics will be:

- Cost per question
- Cost per active learner/month
- GPU utilization
- Tokens/request
- Peak concurrency

13.Evaluation Framework

Evaluation should be education-specific and reviewed with Kenyan teachers.

Evaluation ownership

The dataset should be created from:

- Kenyan curriculum content
- Teacher-authored questions
- Representative student questions
- English and Kiswahili examples
- Different grades and subjects
- Follow-up questions
- Reasoning questions
- Out-of-curriculum questions

The evaluation set should cover questions across relevant Kenyan curriculum subjects and class levels, in both English and Kiswahili.

The initial dataset size will be agreed based on curriculum coverage and teacher-review capacity, with sufficient examples in each language to compare candidate models fairly

14.Evaluation Metrics and Acceptance Gates

Each Candidate system should be evaluated on:

Metric	Initial target
Answer correctness	≥ 90%
Curriculum grounding	≥ 95%
Grade-level appropriateness	≥ 90%
Hallucination rate	≤ 3%
P95 latency	Defined from product requirement
Retry/failure rate	Defined from pilot
Cost/learner/month	Within agreed deployment budget

These are initial proposed gates, not fixed assumptions. They should be agreed with the education/product team before model selection.

A model or architecture change should not be considered production-ready unless it passes the agreed gates.

15.Observability

Each AI request should be traceable through the pipeline:

Question → Classification → Retrieved Content → Prompt → Model → Validation → Final Response

Important metrics:

- Latency
- Tokens
- Model used
- Retrieval results
- Retry count
- Validation failures
- Estimated inference cost
- User feedback

This will make is possible to identify whether a poor answer was caused by Retrieval, prompt, Model, Context or validation.

16. MVP Scope

Build:

- Kenyan Curriculum ingestion
- English + Kiswahili evaluation
- Curriculum ingestion
- RAG
- Class/subject context
- Hybrid retrieval
- Conversational Q&A
- Open-source LLM inference
- AI orchestration
- Image Input
- Edge connectivity handling
- Response validation
- Basic safety
- Evaluation
- Observability

Defer:

Fine-tuning • Advanced personalization • Adaptive learning • Voice • Teacher analytics • Complex autonomous agents

The Technical foundation should be:

RAG + AI Orchestration/Harness + Open-Source LLM + Response Validation + Evaluation

The architecture should remain modular so that model, infrastructure and retrieval components can be changed based on measured results rather than locked in prematurely.

WBS for the approach paper:

I have broken the proposed approach into workstreams, deliverables and estimated timelines. The estimates are preliminary since the overall project duration and implementation scope haven't been finalized.

Workstream	Main deliverable	Indicative duration
Scope and requirements	MVP scope, requirements, acceptance criteria	2–4 days
Evaluation and model benchmarking	English/Kiswahili evaluation & model recommendation	4–5 days
Curriculum ingestion and RAG	Curriculum retrieval pipeline	5–7 days
AI assistant workflow	Query rewriting, generation, validation	4–6 days
Hybrid deployment	Edge/cloud integration and connectivity handling	5–8 days
Image-based questions	Image processing and integration, if in MVP	3–5 days
Observability and cost measurement	Tracing, metrics, and cost reporting	2–4 days
Integration and testing	Quality, connectivity, and load testing	4–5 days
Pilot and acceptance	Teacher review, results, and issue tracking	3–5 days